CoE-XII

Aakash
Medical | IIT-JEE | Foundations
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NCERT BOOSTER TEST SERIES

Test No. –04

Topics covered in various subjects:

Physics : Mechanical Properties of Fluids, Thermal Properties of Matter, Thermodynamics, Kinetic theory

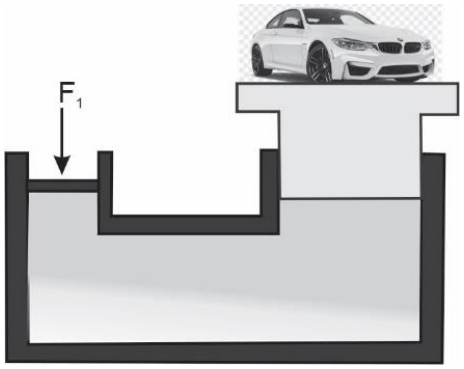
Chemistry: Hydrogen, The *s*-Block elements, The *p*-Block Elements, Organic Chemistry: Some Basic Principles and Techniques

Botany : Mineral Nutrition, Photosynthesis in Higher Plants, Respiration in Plants, Plant Growth and Development, Reproduction in Organisms and Sexual Reproduction in Flowering Plants

Zoology : Chemical Coordination and Integration, Animal Kingdom, **Structural Organisation in Animals:** Animal Morphology of Cockroach, Earthworm, Frog

Choose the correct answer:

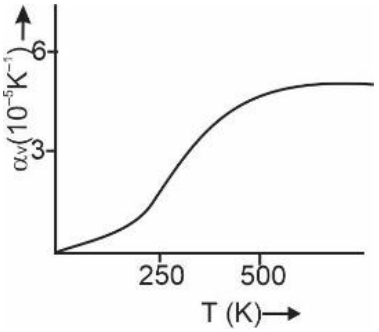
1. Density of Ethyl alcohol at STP is
   (1) 1.03×10^3 kg/m^3 (2) 0.806×10^3 kg/m^3
   (3) 1.06×10^3 kg/m^3 (4) 9.0×10^{-2} kg/m^3
2. The pressure exerted by the atmosphere at sea level is
   (1) 2.013×10^5 Pa (2) 4.013×10^5 Pa
   (3) 1.013×10^5 Pa (4) 5.012×10^5 Pa
3. The two thigh bones (femurs) each cross sectional area 15 cm^2 support the upper part of human body of mass 60 Kg. The average pressure sustained by the femurs will be
   (1) 4×10^5 N/m^2 (2) 2×10^5 N/m^2
   (3) 1×10^5 N/m^2 (4) 5×10^4 N/m^2
4. Which among the following is not a pressure measuring device?
   (1) The mercury barometer
   (2) The open tube manometer
   (3) Hydraulic machine
   (4) Both (2) & (3)
5. Which among the following device does not work on the base of Pascal's law?
   (1) Hydraulic brakes
   (2) The open tube manometer
   (3) Venturi meter
   (4) Both (2) and (3)
6. The gauge pressure at the depth of 500 m in an ocean will be (The density of sea water is 1.03×10^3 kg/m^3 and g = 10 m/s^2)
   (1) 51.5×10^5 Pa (2) 100.3 Pa
   (3) 104.01×10^5 Pa (4) 4.12×10^5 Pa
7. In a car lift compressed air exerts a force F_1 on a small piston having a radius of 10 cm. This pressure is transmitted to a second piston of radius 20 cm as shown in figure. If the mass of the car to be lifted is 1500 kg. The value of F_1 will be

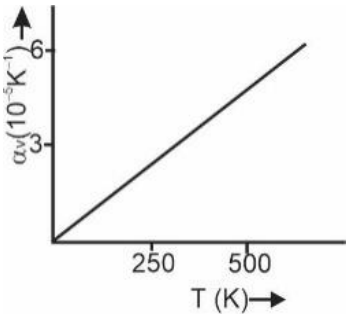


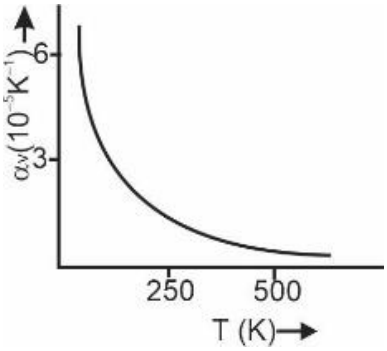
   (1) 1.5×10^4 N (2) 1.5×10^7 N
   (3) 3.75×10^3 N (4) 7.5×10^4 N

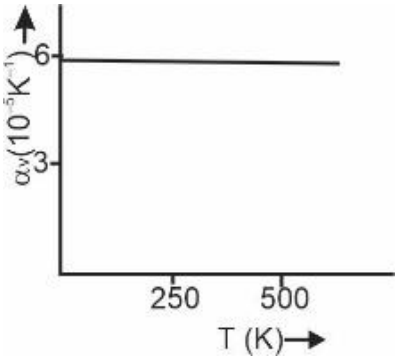
8. The equation of continuity is based on
   (1) Conservation of energy
   (2) Conservation of linear momentum
   (3) Conservation of angular momentum
   (4) Conservation of mass
9. Choose the correct statement among the following
   (1) The blood pressure in humans is greater at the feet than at the brain
   (2) Hydrostatic pressure is a scalar quantity
   (3) Atmospheric pressure at a height of about 6 km decreases to nearly half of its value at the sea level
   (4) All of these
10. Choose the incorrect statement among the following
   (1) The angle of contact of mercury with glass is obtuse, while that of water with glass is acute
   (2) Water on a clean glass surface tends to spread out while mercury on the same surface tends to form drops
   (3) Surface tension of a liquid is dependent of the area of the surface
   (4) Water with detergent dissolved in it should have small angles of contact
11. The viscosity of the glycerine at 20°C temperature in (mPI) is
   (1) 34 (2) 200
   (3) 830 (4) 0.3
12. The flow of blood in a large artery of an anesthetised dog is diverted through a venturi meter. The wider part of the meter has cross-sectional area equal to that of the artery. A = 10 mm^2. The narrower part has an area *a* = 5mm^2. The pressure drop in the artery is 96 Pa. The speed of the blood in the artery will be
   (1) 0.123 m/s (2) 0.246 m/s
   (3) 0.646 m/s (4) 0.323 m/s
13. The ratio of excess pressure inside the air bubble to a liquid drop having same surface tension and radius will be
   (1) 2:1 (2) 4:1
   (3) 1:1 (4) 1:4
14. **Statement (A):** Heat is the form of energy transferred between two (or more) systems or a system and its surroundings by virtue of temperature difference.

   Statement (B): The SI unit of heat energy transferred is expressed in joule(J) while SI unit of temperature is degree Celsius.
   (1) Only statement (A) is correct
   (2) Only statement (B) is correct
   (3) Statement (A) and (B) both are correct
   (4) Neither statement (A) nor statement (B) are correct
15. The value of coefficient of volume expansion of the paraffin in (k^{-1}) is
   (1) 2.2×10^{-5} (2) 18.2×10^{-5}
   (3) 58.8×10^{-5} (4) 110×10^{-5}
16. Choose the correct graph of the coefficient of volume expansion of copper.

(1)


(2)


(3)


(4)


17. Which among the following substance has maximum specific heat capacity at room temperature and atmospheric pressure?

(1) Mercury (2) Ice

(3) Water (4) Kerosene

18. A sphere of aluminium of 0.5 kg is placed for sufficient time in a vessel containing boiling water. So that the sphere is at 100°C. It is then immediately transferred to 0.20 kg copper calorimeter containing 0.30 kg water at 20°C. The temperature of water rises and attains a steady state at 25°C. The specific heat capacity of aluminium will be (S_{cu} = 386 J $kg^{-1}k^{-1}$ and S_w = 4180 J $kg^{-1}k^{-1}$)

(1) 887.5 J Kg^{-1} K^{-1} (2) 177.49 J Kg^{-1} K^{-1}

(3) 1965 J Kg^{-1} K^{-1} (4) 2118 J Kg^{-1} K^{-1}

19. The phenomenon of refreezing of water above the wire when wire having load passing through a slab of ice is called

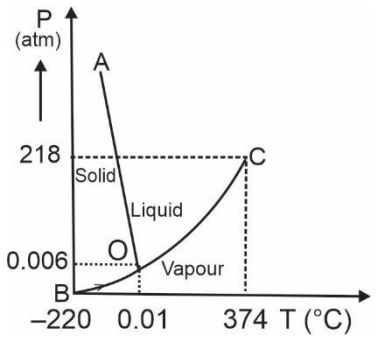
(1) Vaporisation

(2) Sublimation

(3) Regelation

(4) Fusion

20. A graph between the temperature and pressure (P) for water is as shown in figure. The (AO) curve represents



(1) Sublimation

(2) Vaporisation

(3) Fusion

(4) Regelation

21. The rate of flow of heat at steady state through the conductor is inversely proportional to

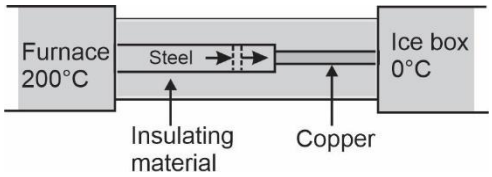
(1) Thermal conductivity

(2) Area of cross-section

(3) Temperature gradient

(4) Differential length of the conductor

22. The temperature of the steel-copper junction in the steady state of the system as shown in figure will be (length of the steel rod = 20.0 cm and length of the copper rod = 10.0 cm, temperature of furnace = 200°C, temperature of the other end = 0°C. The area of cross section of the steel rod is twice that of the copper rod, thermal conductivity of steel = 50.2 $JS^{-1}m^{-1}k^{-1}$, and of copper = 385 $JS^{-1}m^{-1}K^{-1}$)



(1) 30°C (2) 43°C

(3) 20°C (4) 23°C

23. In the trade wind, air returns to the equator and descends at the latitude of

(1) 60° North (2) 30° North

(3) 45° North (4) 60° South

24. In which among the followings modes of heat transfer, gravity play the important role?

(1) Conduction

(2) Radiation

(3) Natural convection

(4) Forced convection

25. The colour of piece of iron heated in a hot flame, first becomes dull red, than radish yellow, and finally white hot, it is explained by the

(1) Stefan-Boltzmann law

(2) Newton's law of cooling

(3) Wein's Displacement law

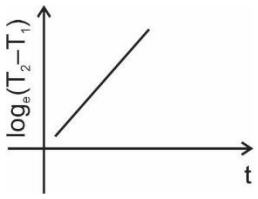
(4) Kirchhoff's law

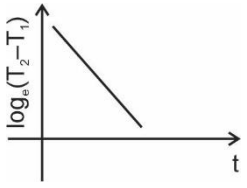
26. Which among the following gases, generally not known as green house gas

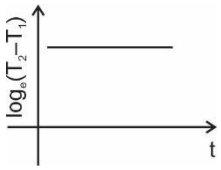
(1) CO_2 (2) CH_4

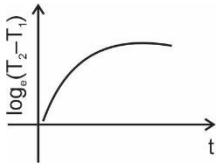
(3) N_2O (4) SO_2

27. Choose the correct graph between $\log_e(T_2-T_1)$ versus time(t), where T_2 is the temperature of water in calorimeter and T_1 is the temperature of hot water in between the double walls in the experimental set up of Newton's law of cooling

(1)


(2)


(3)


(4)


28. Number of state variables by which thermodynamic state specified, is

(1) Three (2) Five

(3) Four (4) Two

29. Which among the following quantities depend on the path between two thermodynamic states?

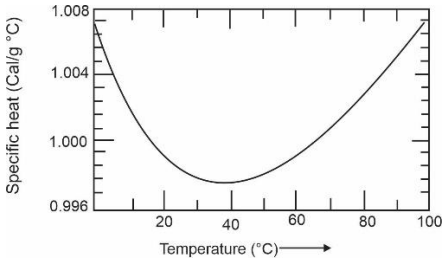
(1) Work done

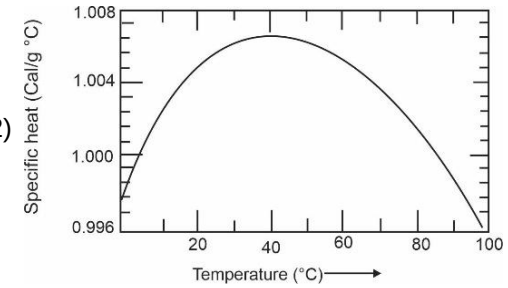
(2) Change in internal energy of gas

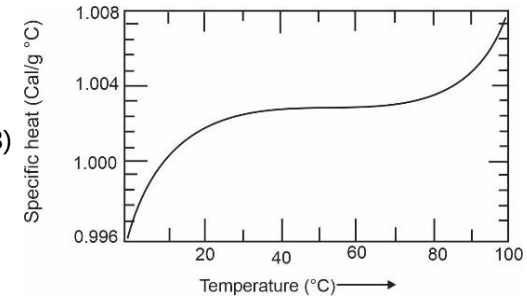
(3) Heat supplied to the system

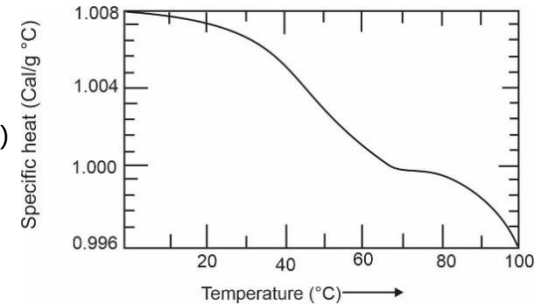
(4) Both (1) & (3)

30. Choose the correct graph of the variation of specific heat capacity of water with temperature

(1)


(2)


(3)


(4)


31. The specific heat of a solid having N atoms, each vibrating about its mean position will be (where R is gas constant)

(1) 2R (2) 3R

(3) R (4) 4R

32. Statement (A): The extensive variables of thermodynamic state, indicate the 'size' of the system.

Statement (B): Pressure and temperature are the intensive variable for a thermodynamic state.

(1) Statement (A) is true

(2) Only statement (B) is true

(3) Both statement (A) and (B) are true

(4) Neither statement (A) nor statement (B) are true

33. In which among the following thermodynamic process, change in internal energy of the gas is zero?
    (1) Isothermal process
    (2) Cyclic process
    (3) Adiabatic process
    (4) Both (1) & (2)

34. If triple point of neon is 24.57 K, then triple point of neon at the Fahrenheit scale will be
    (1) – 415.44 °F (2) – 248.58 °F
    (3) – 56.60 °F (4) – 348.58 °F

35. If the coefficient of volume expansion of brass is $6 \times 10^{-5}\ K^{-1}$, then fractional change in its density for a 60°C rise in temperature will be
    (1) 3.6×10^{-5} (2) 3.6×10^{-3}
    (3) 10^{-6} (4) 2×10^{6}

36. **Statement (A):** No process is possible whose sole result is the absorption of heat from a reservoir and the complete conversion of heat into work.
    Statement (B): No process is possible whose sole result is the transfer of heat from a colder object to a hotter object.
    (1) Only Statement (A) is correct
    (2) Only Statement (B) is correct
    (3) Both Statement (A) and (B) are correct
    (4) Neither Statement (A) Nor statement (B) are correct

37. Choose the incorrect statement among the following
    (1) Efficiency of heat engine never exceeds 1
    (2) The coefficient of performance of a refrigerator can not be infinite
    (3) The coefficient of performance of a refrigerator never exceeds 1
    (4) Both (2) & (3)

38. The cylinders A and B of equal capacity are connected to each other via a stopcock. A contains a gas at standard temperature and pressure. B is completely evacuated. The entire system is thermally insulated. The stop- cock is suddenly opened. Choose the incorrect statement
    (1) The final pressure inside the gas A and B is 1 atm
    (2) The change in internal energy of the gas is zero
    (3) The change in temperature of the gas is zero
    (4) The final pressure inside A and B is 0.5 atm

39. Choose the correct statement among the following
    (1) Real gases approach ideal gas behaviour at low pressure and high temperature
    (2) Real gases approach ideal gas behaviour at high pressure and high temperature
    (3) Real gases approach ideal gas behaviour at high pressure and low temperature
    (4) Real gases approach ideal gas behaviour at low pressure and low temperature

40. A flask contains argon and chlorine in the ratio of 4:1 by mass. The temperature of the mixture is 47°C. The ratio of the average kinetic energy per mole of the two gases will be
    (1) 2:1 (2) 4:1
    (3) 1:1 (4) 8:1

41. A vessel contains two non-reactive gases: neon(monatomic) and oxygen(diatomic): the ratio of their partial pressure is 5:3. The ratio of number of molecules is
    (1) 2:3 (2) 1:1
    (3) 5:3 (4) 1:2

42. Which among the following gas has only translational degree of freedom
    (1) Argon (2) Oxygen
    (3) Nitrogen (4) Carbon dioxide

43. A cylinder of fixed capacity of 67.2 liters helium gas at standard temperature and pressure. The amount of heat required to raise the temperature of the gas in the cylinder by 30.0°C, will be ($R = 8.31\ J\ mol^{-1}K^{-1}$)
    (1) 374 J (2) 1121.85 J
    (3) 247 J (4) 474 J

44. The average distance covered by a molecule between two successive collision is (symbols have their usual meaning)
    (1) $\frac{1}{\sqrt{2} n \pi d}$ (2) $\frac{1}{\sqrt{2} n \pi d^2}$
    (3) $\frac{1}{2 n \pi d}$ (4) $\frac{1}{2 n \pi d^2}$

45. If a diatomic molecule is not rigid but has in addition a vibrational mode then the value of $\frac{C_P}{C_v}$ will be
    (1) $\frac{7}{2}$ (2) $\frac{9}{2}$
    (3) $\frac{7}{5}$ (4) $\frac{9}{7}$

CHEMISTRY

46. In water-gas shift reaction, the products formed are
(1) CO_2 and H_2O (2) CO_2 and H_2
(3) CH_4 and CO_2 (4) H_2 and C_3O_2

47. Incorrect statement among the following is
(1) Boiling point of deuterium is higher than hydrogen
(2) Tritium has two neutrons in the nucleus
(3) Tritium is radioactive and emits low energy β^- particles
(4) Enthalpy of fusion of H_2 is greater than D_2

48. When carbon monoxide reacts with hydrogen in presence of cobalt catalyst then the organic product obtained is
(1) Methanol (2) Methanal
(3) Methanoic acid (4) Glyoxal

49. In Clark's method the chemical species added to hard water to remove its temporary hardness is
(1) NaOH (2) $Ca(OH)_2$
(3) $NaHCO_3$ (4) $CaCl_2$

50. Moles of H_2O_2 required to react completely with one mole of $KMnO_4$ in alkaline medium is
(1) $\frac{5}{2}$ (2) $\frac{3}{2}$
(3) 5 (4) $\frac{5}{4}$

51. Dihedral angle of H_2O_2 structure in gas phase is
(1) 90.2° (2) 95.4°
(3) 105° (4) 111.5°

52. When CaC_2 and Al_4C_3 react with heavy water separately then the products obtained respectively are
(1) C_2D_2 and C_3D_8 (2) C_3D_6 and CD_4
(3) C_2D_2 and CD_4 (4) CD_4 and C_3D_8

53. The alkali metal which gives crimson red colour to the flame test is
(1) Na (2) K
(3) Cs (4) Li

54. Among the given alkali metals the one having lowest density is
(1) Rb (2) Cs
(3) Na (4) K

55. Consider the following statements
(a) Melting point of NaF is greater than NaCl
(b) LiCl is soluble in pyridine
(c) Lithium forms ethynide on reaction with ethyne
The correct statements are
(1) (a) & (b) only (2) (b) & (c) only
(3) (a), (b) & (c) (4) (a) & (c) only

56. Thermally most stable compound among the following is
(1) $LiCO_3$ (2) $NaCO_3$
(3) KCO_3 (4) $CsCO_3$

57. The compound which on heating does not liberates NO_2 gas is
(1) $Be(NO_3)_2$ (2) $Ba(NO_3)_2$
(3) KNO_3 (4) $LiNO_3$

58. During preparation of Sodium hydroxide using Castner-Kellner cell the cathode used in the electrolysis process is
(1) Zn (2) Hg
(3) Graphite (4) Pb

59. Incorrect statement among the following is
(1) Enzymes that utilize ATP in phosphate transfer require calcium as cofactor
(2) Calcium concentration in plasma is maintained by calcitonin and parathyroid hormone
(3) Calcium pays important role in blood coagulation
(4) Magnesium is present in chlorophyll molecule

60. Correct order of first ionisation energy of B, In, Ga and Al is
(1) B > Al > Ga > In (2) B > Ga > Al > In
(3) B > In > Ga > Al (4) Al > B > Ga > In

61. Number of two centre-two electron bonds present in B_2H_6 is
(1) 2 (2) 4
(3) 6 (4) 0

62. Consider the following statements
(i) Aqueous solution of borax is alkaline in nature
(ii) $Co(BO_2)_2$ is blue coloured compound
(iii) Boric acid is formed on acidifying aqueous solution of borax
(iv) Boric acid is monobasic protonic acid
The correct statements are
(1) (i), (ii) & (iii) only (2) (i) & (iii) only
(3) (i), (ii), (iii) & (iv) (4) (i), (iii) & (iv) only

63. In which pair, both the species are acidic in nature?

(1) SnO_2 and PbO_2 (2) GeO_2 and PbO

(3) SnO_2 and GeO (4) GeO_2 and SiO_2

64. Incorrect statement among the following is

(1) Charcoal is obtained by heating wood at high temperature in absence of air

(2) Carbon black is obtained by burning hydrocarbons in limited supply of air

(3) Carbon in C_{60} is sp^2 hybridised

(4) C_{60} contains twenty six-membered rings and ten five membered ring

65. Which metal oxide(s) can be reduced to their respective metal(s) using CO as reducing agent?

(1) Al_2O_3 (2) CaO

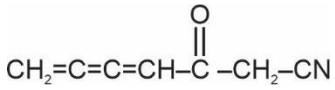
(3) Fe_2O_3 (4) Both (1) & (3)

66. Which compound on hydrolysis followed by condensation polymerisation yields straight chain polymer?

(1) $MeSiCl_3$ (2) Me_3SiCl

(3) Me_2SiCl_2 (4) Me_4Si

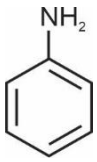
67. Number of sp hybridised carbon in the given molecule is

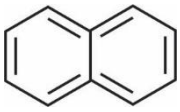


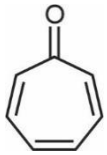
(1) 5 (2) 4

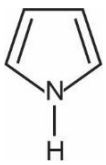
(3) 2 (4) 3

68. Which among the following is a heterocyclic aromatic compound?

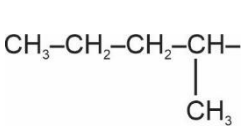
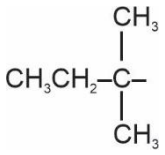
(1) 

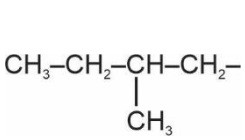
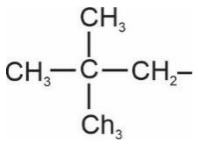
(2) 

(3) 

(4) 

69. Which among the following is neopentyl group?

(1)  (2) 

(3)  (4) 

70. IUPAC name of the given compound is



(1) 1, 1-Diethyl-3-methyl cycloheptane

(2) 3, 3-Diethyl-1-methyl cycloheptane

(3) 6, 6-Diethyl-1-methyl cycloheptane

(4) 1, 1-Diethyl-6-methyl cycloheptane

71. Correct order of decreasing priority sequence of given functional groups in IUPAC nomenclature is

$–SO_3H$, –COOH, $–COOCH_3$, $–CONH_2$

(1) $–CONH_2 > –COOCH_3 > –COOH > –SO_3H$

(2) $–SO_3H > –COOH > –COOCH_3 > CONH_2$

(3) $–COOH > –SO_3H > –COOCH_3 –CONH_2$

(4) $–COOH > –SO_3H > –CONH_2 > –COOCH_3$

72. $C_2H_5OC_3H_7$ and $CH_3OC_4H_9$ are

(1) Chain isomers

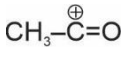
(2) Position isomers

(3) Metamers

(4) Functional group isomers

73. The species which is not an electrophile is

(1) BF_3 (2) 

(3) $(CH_3)_3N$ (4) 

74. Incorrect statement among the following is

(1) The electron displacements due to the influence of an atom or a substituent group present in the molecule cause permanent polarisation of bond

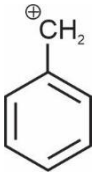
(2) Inductive effect operates through π-bond

(3) Electrometric effect is temporary effect

(4) Resonance structures are hypothetical and individually do no represent any real molecule.

75. In which of the following species hyperconjugation does not occur?

(1) $CH_3\overset{\oplus}{C}H_2$ (2) $CH_3CH=CH_2$

(3) $(CH_3)_3\overset{\oplus}{C}$ (4) 

76. Glycerol can be separated from spent-lye in soap industry by

(1) Differential extraction

(2) Steam distillation

(3) Distillation under reduced pressure

(4) Fractional distillation

77. On treating sodium fusion extract with sodium nitroprusside, appearance of violet colour indicates the presence of

(1) Nitrogen only

(2) Sulphur only

(3) Both nitrogen and sulphur

(4) Phosphorus

78. Kjeldahl method is not applicable to which compound?

(1) Aniline (2) Pyridine

(3) Benzyl amine (4) Triethyl amine

79. In Carius method of estimation of halogen, 1.25 g of an organic compound gave 0.94 g of AgBr. Percentage of bromine in the organic compound is (Atomic mass of Ag = 108 u; Br = 80 u)

(1) 16.5 % (2) 45.6 %

(3) 32 % (4) 64 %

80. Colour of ammonium phosphomolybdate is

(1) Green (2) Red

(3) Blue (4) Yellow

81. Volume strength of 3 % H_2O_2 solution is

(1) 5 V (2) 10 V

(3) 15 V (4) 20 V

82. If E° for Li^+/Li is –3.04 V, for Cl_2/Cl^- is +1.36 V, for I_2/I^- is +0.53, for Ag^+/Ag is +0.79 V and Na^+/Na is –2.71 V then decreasing order of reducing strength of Ag, Li, I^-, Cl^- and Na will be

(1) $I^- > Cl^- > Ag > Na > Li$

(2) $Ag > Na > Li > I^- > Cl^-$

(3) $Li > Na > I^- > Ag > Cl^-$

(4) $Li > Na > Ag > I^- > Cl^-$

83. Which among the following is most soluble in water?

(1) $Mg(OH)_2$ (2) $Ca(OH)_2$

(3) $Ba(OH)_2$ (4) $Sr(OH)_2$

84. Which of the given alkali metals gives hydrated salts?

(1) Li (2) Na

(3) K (4) Cs

85. Correct order of ionic mobilities of given ions in aqueous solution is

Rb^+, K^+, Na^+, Li^+

(1) $Li^+ > Na^+ > K^+ > Rb^+$ (2) $Na^+ > K^+ > Rb^+ > Li^+$

(3) $Rb^+ > Na^+ > K^+ > Li^+$ (4) $Rb^+ > K^+ > Na^+ > Li^+$

86. Thermodynamically the most stable form of carbon is

(1) Coal (2) Graphite

(3) C_{60} (4) Diamond

87. Correct order of relative stability of +3 oxidation state of Tl, In, Ga and Al is

(1) Al < Ga < In < Tl (2) Tl < In < Ga < Al

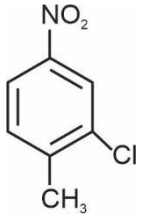
(3) In < Tl < Ga < Al (4) Al < Tl < Ga < In

88. Electron-precise hydrid among the following is

(1) B_2H_6 (2) NH_3

(3) HF (4) CH_4

89. IUPAC name of the given compound is



(1) 2-Chloro-1-methyl-4-nitrobenzene

(2) 1-Chloro-2-methyl-5-nitrobenzene

(3) 4-Methyl-5-chloronitrobenzene

(4) 3-Chloro-4-methylnitrobeneze

90. During estimation of nitrogen present in an organic compound by Kjeldahl's method, the ammonia evolved from 0.8 g of compound was neutralised by 15 ml of 0.5 M H_2SO_4 solution. The percentage of nitrogen in the organic compound is

(1) 56 % (2) 16 %

(3) 26.25 % (4) 15.75 %

Botany

91. The scientist who demonstrated for the first time that plants could be grown to maturity in a defined nutrient solution *i.e.* hydroponics was a

(1) British botanist

(2) German botanist

(3) Amarican physiologist

(4) Russian agriculturist

92. Which of the following is **not** a criterion for essentiality of an element for a plant?

(1) In the absence of the element, plants do not complete their life cycle

(2) The element must be directly involved in the plants metabolism

(3) The requirement of one element is not replaceable by another element

(4) The element must be a component of a plant growth regulator

93. Out of the following elements, which one is absorbed from the soil as mineral nutrition?

(1) Nitrogen (2) Carbon

(3) Oxygen (4) Hydrogen

94. Select the **odd one out** w.r.t. essential elements for plants.

(1) Sulphur (2) Cobalt

(3) Nickel (4) Zinc

95. The essential element that is component of energy-related chemical compound chlorophyll is

(1) Copper (2) Manganese

(3) Magnesium (4) Phosphorus

96. Sulphur is **not** a constituent of

(1) Thiamine (2) Biotin

(3) Cellulose (4) Coenzyme A

97. Translocation of which element in shoot apex is inhibited by manganese?

(1) Calcium (2) Cupper

(3) Potassium (4) Molybdenum

98. Match the following columns and select the **correct** option.

	Column I		**Column II**
a	*Azotobacter*	(i)	Free-living nitrogen-fixing
b	*Nostoc*	(ii)	Free-living nitrogen-fixing and aerobic
c	*Rhodospirillum*	(iii)	Symbiotic nitrogen-fixing
d	*Rhizobium*	(iv)	Free-living nitrogen-fixing and anaerobic

(1) a (iv), b (iii), c (ii), d (i)

(2) a (ii), b (iv), c (i), d (iii)

(3) a (iv), b (i), c (ii), d (iii)

(4) a (ii), b (i), c (iv), d (iii)

99. The function of leg-haemoglobin in the root nodules of legume plants is to

(1) Protect nitrogenase from oxygen

(2) Provide nitrogen to nitrogen-fixing microbes

(3) Supply oxygen to nitrogenase

(4) Reduce nitrogen to ammonia

100. Through his experiments, Jan Ingenhousz showed that
   (1) Carbon dioxide is essential for the synthesis of carbohydrates in plants
   (2) Oxygen evolved during photosynthesis comes from water
   (3) Only green parts of plant can synthesize glucose
   (4) Sunlight is essential to the plant process that purifies the air

101. In the stroma of chloroplast, sugar synthesis is dependent on the products of light reaction which are
   (1) ATP and NADPH (2) H^+ and O_2
   (3) $NADH+H^+$ and O_2 (4) CO_2 and H_2O

102. The photosynthetic pigment which gives bright or blue-green colour in the chromatogram is/are
   (1) Xanthophylls (2) Chlorophyll *b*
   (3) Chlorophyll *a* (4) Carotenoids

103. The wavelengths at which there is maximum absorption by chlorophyll *a* are in the regions of
   (1) Green and red spectra
   (2) Blue and red spectra
   (3) Green and yellow spectra
   (4) Blue and green spectra

104. The pigments in light harvesting complex of a photosystem can be
   (a) Chlorophyll *a* (b) Chlorophyll *b*
   (c) Carotenoids (d) Xanthophylls
   (1) (a) and (b) only
   (2) (c) and (d) only
   (3) (b), (c) and (d) only
   (4) All (a), (b), (c) and (d)

105. In Z scheme of light reaction of photosynthesis, the electrons are finally received by
   (1) ADP (2) $NADP^+$
   (3) $NADPH+H^+$ (4) PS II

106. In the chloroplast, water splitting complex is located on the
   (1) Inner side of inner membrane of chloroplast
   (2) Outer surface of stromal lamellae
   (3) Inner side of thylakoid membrane
   (4) Outer surface of thylakoid membrane

107. The non-cyclic photophosphorylation occurs in grana lamellae because the lamellae
   (1) Lack PS II as well as NADP reductase enzyme
   (2) Have only PS I and ATPase enzyme
   (3) Lack chlorophyll *a* but have chlorophyll *b*
   (4) Have both PS I and PS II as well as NADP reductase enzyme

108. Identify the **false** statement regarding the primary CO_2 acceptor molecule in Calvin cycle.
   (1) It is a 2-carbon compound
   (2) It is a ketose sugar
   (3) It has two phosphate groups
   (4) It combines with CO_2 to from C_3-acid

109. The most crucial step of Calvin cycle is
   (1) Carboxylation (2) Oxidation
   (3) Regeneration (4) Reduction

110. Which enzyme helps in the primary fixation of CO_2 with PEP in C_4-cycle?
   (1) RuBisCO
   (2) Pyruvate kinase
   (3) Phosphoenol pyruvate carboxylase
   (4) Pyruvate dehydrogenase

111. The property of RuBisCO which is responsible for photorespiration in C_3-plants is its activity as
   (1) Carboxylase (2) Oxygenase
   (3) Dehydrogenase (4) Decarboxylase

112. Carbon dioxide is the major limiting factor for photosynthesis because
   (1) It is produced during respiration
   (2) It is available only in the form of gas
   (3) Its concentration in atmosphere is very low
   (4) It is the major source of carbon for glucose synthesis

113. Breaking of C-C bonds of complex compounds and release of energy within a cell is called
   (1) Assimilation (2) Respiration
   (3) Dissociation (4) Fermentation

114. Which of the following steps does not occur in glycolysis?
   (1) 2-phophoglycerate $\rightarrow$ Phosphoenol pyruvate
   (2) 1,3-bisphosphoglyceric acid $\rightarrow$3-phosphoglyceric acid
   (3) Fructose-1, 6 - bisphosphate $\rightarrow$Glyceraldehyde-3-phosphate
   (4) Pyruvic acid $\rightarrow$ Acetyl coenzyme A

115. how many molecules of ATP is/are directly synthesised through glycolysis from one molecule of glucose?

(1) Two (2) One

(3) Four (4) Six

116. What percent of energy in glucose is released in alcoholic fermentation?

(1) About 13% (2) More than 36%

(3) About 20% (4) Less than 7%

117. Identify the conversion step during which the substrate level phosphorylation occurs in the citric acid cycle.

(1) Oxaloacetic acid $\rightarrow$ Citric acid

(2) Succinic acid $\rightarrow$ Malic acid

(3) Succinyl CoA $\rightarrow$ Succinic acid

(4) Citric acid $\rightarrow$ α-ketoglutaric acid

118. Match the following columns and select the **correct** option w.r.t. different complexes in ETS.

	Column I	**Column II**
a	Complex III	(i) Succinate dehydrogenase
b	Complex I	(ii) NADH dehydrogenase
c	Complex IV	(iii) Cytochrome bc_1
d	Complex II	(iv) Cytochrome c oxidase

(1) a (ii), b (iv), c (i), d (iii)

(2) a (iii), b (ii), c (i), d (iv)

(3) a (iii), b (ii), c (iv), d (i)

(4) a (iii), b (i), c (iv), d (ii)

119. Glycerol enters the respiratory pathway after being converted to

(1) 3-phosphoglyceraldehyde

(2) Pyruvic acid

(3) Acetyl CoA

(4) Fructose-1, 6-bisphosphate

120. A respiratory substrate is oxidised as shown in the equation below:

$C_{51}\ H_{98}\ O_6 + O_2 \rightarrow CO_2 + H_2O$ + Energy

What would be the respiratory quotient of this substrate?

(1) 0.7 (2) 0.9

(3) 1.33 (4) 1.0

121. Plants retain the capacity for unlimited growth throughout their life. This ability of plants is due to

(1) Lack of complex tissues

(2) Presence of meristems

(3) Presence of permanent tissues

(4) Absence of organ system of organisation

122. The growth curve for living organisms growing in a natural environment is

(1) Linear (2) L-shaped

(3) Bell-shaped (4) Sigmoid

123. Cork cambium is an example of

(1) Redifferentiated tissue

(2) Dedifferentiated tissue

(3) Differentiated tissue

(4) Complex permanent tissue

124. The ability of plants to follow different pathways in response to environment or phases of life to form different kinds of structures is called

(1) Adaptability (2) Plasticity

(3) Elasticity (4) Susceptibility

125. Select the **incorrect** match from the following

(1) Auxin – Induces parthenocarpy

(2) Gibberellin – Helps to overcome the apical dominance

(3) Cytokinin – Helps to produce chloroplasts in leaves

(4) Ethylene – A gaseous plant growth regulator

126. In most situations, ABA acts as antagonist to

(1) Cytokinins (2) Auxins

(3) Ethylene (4) Gibberellins

127. Read the following statements and choose the option which is true for them.

Statement A: Vernalisation refers to the promotion of flowering by a period of low temperature.

Statement B: Vernalisation promotes precocious reproductive development much earlier in the growing season.

(1) Only statement A is correct

(2) Only statement B is correct

(3) Both the statements are correct

(4) Both the statements are incorrect

128. The **correct** sequence w.r.t. the life spans of different organisms is
(1) Rice plant > Rose plant > Banana plant
(2) Crow > Parrot > Banyan tree
(3) Tortoise > Crocodile > Crow
(4) Butterfly > Rice plant > Fruit fly

129. Soprulation can be seen
(1) Only in prokaryotes
(2) In both unicellular and multicellular organisms
(3) Only in unicellular organisms
(4) Only in the organisms which have cell wall

130. Vegetative propagation through the formation of offset is seen in
(1) *Agave* (2) *Bryophyllum*
(3) Water hyacinth (4) Ginger

131. Cells of which structure of the anther undergo meiotic division to form microspore tetrads?
(1) Tapetum
(2) Nucellus
(3) Endothecium
(4) Sporogenous tissue

132. Select the **correct** match from the following w.r.t. angiospermic plants.
(1) *Michelia* – Syncarpous pistil
(2) Mature embryo sac - 8-nucleate stage
(3) Synergids – Situated towards chalazal end
(4) Egg apparatus – One egg cell and two polar nuclei

133. Which of the following statements is/are **incorrect**?
(a) Geitonogamy does not occur in the plants having bisexual flowers.
(b) Xenogamy is seen only in dioecious plants.
(c) Cross pollination is not possible in plants having only cleistogamous flowers.
(1) (b) only (2) (a) and (b) only
(3) (a) only (4) All (a), (b) and (c)

134. In the embryo of the grass family, which structure is called scutellum?
(1) Cotyledon (2) Hypocotyl
(3) Plumule (4) Radicle

135. Polyembryony is the case in which
(1) More than one seed are formed from one ovule
(2) There is the occurrence of more than one embryo in a seed
(3) More than one seed is formed in one ovary
(4) Many embryos are developed out of the ovule

Zoology

136. Chemical coordination and integration provides
(1) Point to point contact
(2) Rapid coordination
(3) Continuous regulation of cellular functions along with neural system
(4) Innervation to all cells of the body

137. Which of the following is **incorrect** w.r.t. hormones?
(1) Non-nutrient chemicals
(2) Intercellular messengers
(3) Produced in trace amounts
(4) Hormones are secreted by the organised endocrine glands only

138. Choose the **odd** one out w.r.t. source gland
(1) PRL (2) ACTH
(3) GnRH (4) GH

139. Which part of pituitary is under the direct neural regulation of the hypothalamus?
(1) Neurohypophysis (2) Adenohypophysis
(3) Pars distalis (4) Pars intermedia

140. Match the column-I with column-II and choose the **correct** option

Column-I	**Column-II**
(a) Hypersecretion of GH in middle age	(i) Growth of mammary glands and synthesis of milk
(b) Prolactin	(ii) Spermatogenesis
(c) FSH and androgens	(iii) Ovulation
(d) LH	(iv) Acromegaly
	(v) Gigantism

(1) a(iv), (b)(iii), c(ii), d(i)
(2) a(v), (b)(i), c(ii), d(iii)
(3) a(iv), (b)(i), c(ii), d(iii)
(4) a(v), (b)(iii), c(ii), d(i)

141. Which of the following hormone maintains corpus luteum?
(1) FSH (2) LH
(3) GnRH (4) GH

142. Which of the following is **not** the function of hormone which is secreted by gland located on the dorsal side of forebrain?
(1) Maintains diurnal rhythm
(2) Maintains body temperature
(3) Influences pigmentation
(4) Synthesis and secretion of glucocorticoids

143. Select the hormone which is secreted by the thyroid gland but **not** by the follicular cells.
(1) PTH (2) TCT
(3) T3 (4) T4

144. Complete the analogy
Insulin: Hypoglycaemia::Parathyroid hormone: _____.
(1) Hypocalcaemia
(2) Hyperglycaemia
(3) Hypercalcemia
(4) Hypothyroidism

145. Enlargement of thyroid gland, protrusion of the eyeballs, increased BMR and weight loss are the symptoms of
(1) Simple goitre (2) Cretinism
(3) Graves' disease (4) Addison's disease

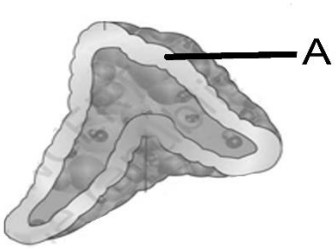
146. Read the following statements and choose the **correct** option.
Statement A: Thyroid hormones influence water and electrolyte balance in the body.
Statement B: Thyroid gland secretes a protein hormone which regulates the blood calcium level.
(1) Both A and B statements are correct
(2) Only statements A is correct
(3) Only statements B is correct
(4) Both statements A and B are incorrect

147. Match the column-I with column-II and choose the option with all **correct** match.

Column-I	Column-II
(a) Thymus	(i) Anterior part of each kidney
(b) Parathyroid glands	(ii) Either side of the trachea
(c) Thyroid gland	(iii) Back side of the thyroid gland
(d) Adrenal gland	(iv) Between lungs and behind sternum

(1) a-iv, b-iii, c-ii,d-i
(2) a-iii, b-iv, c-i,d-ii
(3) a-ii, b-i,c-iii, d-iv
(4) a-i, b-ii, c-iii, d-iv

148. Which of the following is **not** the function of hormones secreted by the labelled part A in the given below diagram?



(1) Maintain the cardio-vascular system and kidney functions.
(2) Reabsorption of Na^+ and water and excretion of K^+ and phosphate
(3) Growth of axial hair, pubic hair and facial hair during puberty
(4) Pupilary dilation, piloerection and sweating.

149. Which hormone inhibits gastric secretion and its motility?
(1) Gastrin (2) CCK
(3) ANF (4) GIP

150. Choose the **mismatch**
(1) Epinephrine – Aminoacid derivative
(2) Thyroxine – Generate second messenger
(3) FSH – Receptors on ovarian cell membrane
(4) Estrogen – Regulate female sexual behaviour

151. Read the following statements and choose the option with **correct** statements only.

(a) All members of animalia are multicellular.

(b) Some division of labour is present among the cells in the animals exhibiting cellular level of organisation.

(c) Digestive system is complete in Platyhelminths.

(d) In closed circulatory system, blood flows in blood vessels of varying diameter.

(1) a, b, d (2) a, c, d

(3) a and d only (4) b and c only

152. Read the following statements and choose the option which represents all true and false statements **correctly**.

(a) All sponges are aquatic.

(b) All members of the phylum Porifera are asymmetrical.

(c) Sponges possess water vascular system.

(d) Sponges are dioecious.

	(a)	**(b)**	**(c)**	**(d)**
(1)	T	T	F	T
(2)	T	F	F	T
(3)	F	F	T	F
(4)	T	F	F	F

153. Skeleton is present in all of the following animals **except**

(1) *Sycon* (2) *Echinus*

(3) *Aplysia* (4) *Hydra*

154. Select the animal which is metamerically segmented and dioecious

(1) *Pheretima* (2) *Nereis*

(3) *Ancylostoma* (4) *Hirudinaria*

155. Choose the **mismatch**

(1) *Apis* – Malpighian tubules

(2) *Nereis* – Nephridia

(3) *Cucumaria* – Protonephridia

(4) *Wuchereria* – Excretory tube

156. The space between the __A__ and the __B__ is called the mantle cavity. Choose the option which **correctly** fill the blanks A and B respectively

(1) Shell, mantle

(2) Mantle, Radula

(3) Hump, mantle

(4) Hump, muscular foot

157. Choose the **odd** one out w.r.t. feather like gills of molluscs

(1) Present in mantle cavity

(2) Respiration

(3) Excretion

(4) Present in visceral hump

158. Ventral heart and excretion through kidneys are features of

(1) All chordates

(2) Protochordates only

(3) Jawed vertebrates only

(4) All vertebrates

159. Read the given below statements and choose the **correct** option.

Statement A: Notochord is dorsal, single and hollow.

Statement B: Notochord is dorsal to nerve cord.

(1) Both A and B statement are correct

(2) Only statement A is correct

(3) Only statement B is correct

(4) Both A and B statements are incorrect

160. Match the column-I with column-II and choose the option with all **correct** match

Column-I	**Column-II**
(a) *Petromyzon*	(i) 4 pairs of gills
(b) *Clarias*	(ii) 6–15 pairs of gill slits
(c) *Pavo*	(iii) Gills without operculum
(d) *Scoliodon*	(iv) Lungs with air sacs

(1) a-i, b-iii, c-ii, d-iv

(2) a-ii, b-i, c-iv, d-iii

(3) a-iv, b-iii, c-i, d-ii

(4) a-iii, b-iv, c-ii, d-i

161. Choose the **odd** one out w.r.t. scales

(1) *Hyla* (2) *Myxine*

(3) *Calotes* (4) *Pteropus*

162. Select the feature which is **not** exclusive to mammals.

(1) Hair

(2) Ear pinnae

(3) Homeothermy

(4) Different types of teeth

163. Open circulatory system is present in

(1) *Nereis* (2) *Balanoglossus*

(2) *Pristis* (4) *Amphioxus*

164. External fertilisation with indirect development is present in all of the following **except**

(1) Porifera (2) Ctenophora

(3) Echinodermata (4) Hemichordata

165. How many of the given below features are **true** for round worms?

(a) May be free living, aquatic and terrestrial

(b) Parasite in animals only

(c) Development may be direct or indirect

(d) Dioecious

(1) 4 (2) 2

(3) 3 (4) 1

166. Choose the **odd** one out w.r.t. cockroaches

(1) Size ranges from 2.6 to 7.6 cm only

(2) Nocturnal

(3) Omnivores

(4) *Periplaneta americana* is about 34-53 mm long

167. Dorsal sclerite is known as

(1) Tergum

(2) Sternum

(3) Articular membrane

(4) Arthrodial membrane

168. Read the following statements and choose the option with **correct** statements only.

(a) In *Periplaneta* wings extend beyond the tip of abdomen in females.

(b) Head is at right angle to the longitudinal axis of the body.

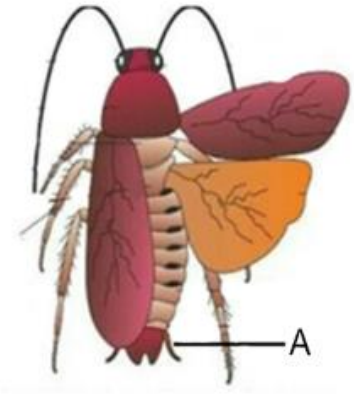
(c) Head shows great mobility in all directions because of flexible neck.

(d) A pair of thread like antennae arise from membranous sockets lying behind the eyes.

(1) a, b, c, d (2) a, b, c only

(3) b, c, d only (4) b and c only

169. Given below is a diagram of cockroach, which of the following is **not** correct for the labelled part **A**.



(1) Jointed filamentous structure

(2) Attached to 9th segment

(3) Present in both sexes

(4) Different from caudal style

170. Mouth parts of cockroach are

(1) Piercing and sucking type

(2) Biting and chewing type

(3) Chewing and lapping type

(4) Sponging type

171. Anterior part of female genital pouch has openings for all **except**

(1) Common oviduct

(2) Spermatheca

(3) Collaterial glands

(4) Ovarioles

172. All of the following are paired structures in cockroach **except**

(1) Collaterial gland

(2) Spermatheca

(3) Testis

(4) Phallic gland

173. Malpighian tubules

(1) Are present at the junction of foregut and midgut

(2) Are lined by glandular and ciliated cells

(3) Absorb uric acid from the haemolymph and excrete out through the hindgut

(4) Are 6–8 blind tubules

174. Select the **correct** match w.r.t. structures and their number in cockroach.
(1) Alary muscles – 10 pairs
(2) Both compound eyes – 2000 ommatidia
(3) One ootheca – 9-10 eggs
(4) Salivary glands – 1 pair

175. Male genital pouch is bounded dorsally by 9th and 10th terga and ventrally by the _____. Choose the option which **correctly** fills the blank.
(1) 9th sternum
(2) 10th sternum
(3) 7th sternum
(4) 8th and 9th sternum

176. Brain of cockroach supplies nerves to
(1) Labrum and labium
(2) Antennae and labium
(3) Compound eyes and labium
(4) Antennae and compounds eyes

177. Choose the **incorrect** statement w.r.t. nervous system of cockroach.
(1) Consists of a series of fused segmentally arranged ganglia
(2) Three pairs of ganglia in the thorax and six pairs in the abdomen
(3) Paired longitudinal connectives on the ventral side
(4) Brain is represented by supra-oesophageal ganglion

178. Match the column-I with column-II and choose the **correct** option

	Column-I	**Column-II**
(a)	Mushroom gland	(i) 2nd - 6th abdominal segment
(b)	Ovaries	(ii) 6th abdominal segment
(c)	Spermathecae	(iii) 6th-7th abdominal segment
(d)	Testes	(iv) 4th-6th abdominal segment

(1) a-iii, b-i, c-ii, d-iv
(2) a-ii, b-iv, c-iii, d-i
(3) a-iv, b-i, c-iii, d-ii
(4) a-i, b-iv, c-ii, d-iii

179. The _____ in male frog acts as urinogenital duct which opens into the cloaca. Choose the option which **correctly** fills the blank.
(1) Urethra
(2) Ureter
(3) Urinary bladder
(4) Uriniferous tubule

180. Circulatory system of which of the following animal has capillaries?
(1) *Pheretima* (2) *Periplaneta*
(3) *Saccoglossus* (4) *Pila*

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